

HTML

HTML (HyperText Markup Language) is the standard language used to create the structure of web pages. Think of it as the **skeleton** of a website—it defines where the headings, paragraphs, and images

Tags

- **<html>** The "Container." It holds the entire website.

```
<html lang="en">
```

- **<head>** The "Brain." It stores hidden info like the title and settings.

```
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="">
  <title>Images</title>
</head>
```

- **<body>** The "Visible Part." Everything you see on the screen goes here.

```
<body>
  Hello World
</body>
```

- **<h1>** to **<h6>** Headings. **<h1>** is the biggest; **<h6>** is the smallest.

```
<h1>Vaibhav</h1>
<h2>Vaibhav</h2>
<h3>Vaibhav</h3>
<h4>Vaibhav</h4>
<h5>Vaibhav</h5>
<h6>Vaibhav</h6>
```


- **** & **** Unordered List. For bullet points.

```
<ul>
  <li>Shoes</li>
  <li>Shirts</li>
  <li>Hats</li>
</ul>
```

- **<input>** Typing Box. Where users type their name or password

```
<input type="text" placeholder="Enter your name">
```

- **<button>** Clickable Button. Does an action when clicked.

```
<button>Click Me!</button>
```

- **<form>** Collection. Groups all inputs and buttons together (like a Login form).

```
<form>
  <input type="email" placeholder="Your Email">
  <button>Join ShopEase</button>
</form>
```

- **<table>** Create Table.
- **<tr>** Table Row (Line).
- **<td>** Table Data (Single box).

```
<table>
  <tr>
    <td>Product</td>
    <td>Price</td>
  </tr>
  <tr>
    <td>Shoes</td>
    <td>$50</td>
  </tr>
</table>
```

- **<header>** The Top Banner for your logo and menu.
- **<nav>** The Map where your navigation links live.
- **<main>** The Heart of the page (the most important content).
- **<footer>** The Bottom Info for contact details and links.
- **<section>** A Themed Block to group related content together.

```
<header>ShopEase Logo</header>
  <nav>Home | Shop | Cart</nav>
<main>
  <section>
    <h2>Featured Products</h2>
    <p>Check out our new shoes!</p>
  </section>
</main>
<footer>Contact us: help@shopease.com</footer>
```

- **
** Break the line.

```
Hello!
<br>
Welcome to ShopEase.
```

More Tags

<https://developer.mozilla.org/en-US/docs/Web/HTML/Reference/Elements>

SEO (Search Engine Optimization)

It's like putting a big, clear sign on your shop so people can find it.

- <title>** The name of your page (shows up on the Google tab).
- <meta name="description">** A short summary that Google shows under your link.
- <h1>** Tells Google, "This is the most important topic on this page."
- alt attribute in **** Tells Google what is in the photo (since Google can't "see" pictures like we do).

Block Elements (The Space Hogs)

These elements are like big bricks. They always start on a new line and take up the full width available (from left to right).

Behavior: They don't like sharing their "line" with anyone else.

Examples: `<div>`, `<h1>`, `<p>`, `<section>`, `<ul>`, etc...

Inline Elements (The Team Players)

These are like stickers. They only take as much width as they need and stay on the same line with other elements.

Behavior: They sit side-by-side with their neighbors until there's no room left.

Examples: `<span>`, `<a>`, `<img>`, `<strong>`, etc...

ID (The Unique Name)

A name that is unique to only one element on the entire page.

Eg - Your Fingerprint or your Social Security ID.

No two people have the same one.

Symbol in CSS: `#` (like `#header`)

```
<div id="main-logo">ShopEase</div>
```

```
#main-logo {  
  color: aquamarine;  
}
```

Classes (The Team Jersey)

A name that can be used for many elements at the same time.

Eg - A School Uniform. Everyone in the same class wears the same blue shirt.

Symbol in CSS: . (like `.blue-button`)

```
<button class="buy-btn">Shoes</button>
<button class="buy-btn">Shirts</button>
```

```
.buy-btn {
  color: bisque;
}
```

Attribute

An **Attribute** is simply **Extra Info** you add to a tag.

It tells the tag "how to behave" or "where to go."

Always inside the opening tag < >.

```

```

src Tells the browser where the photo is.

alt Tells the browser what the photo is.

Non-Semantic (<div>)

An Empty Box. The browser has no idea if it's a menu, a footer, or a title.

Semantic (<header>, <nav>)

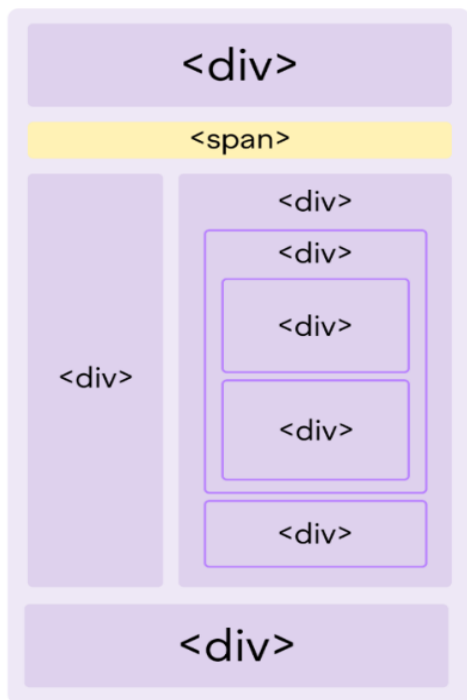
A Labeled Box. The browser knows exactly what its purpose is.

It helps your site show up higher in search results (**SEO**).

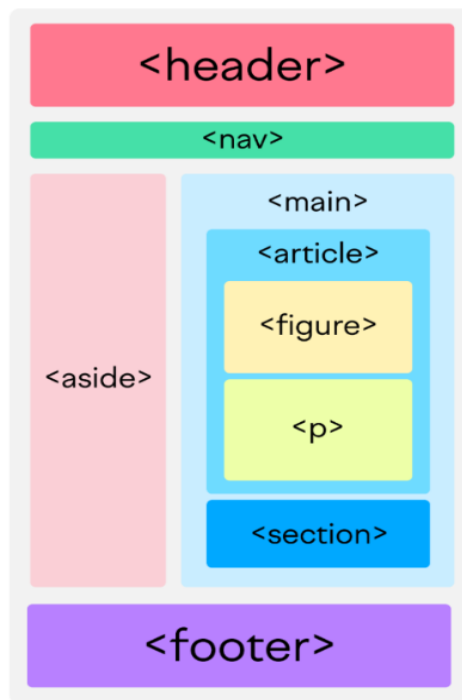
It makes your code look like it was written by a **Pro**.

What Is Semantic HTML?

Non-Semantic HTML



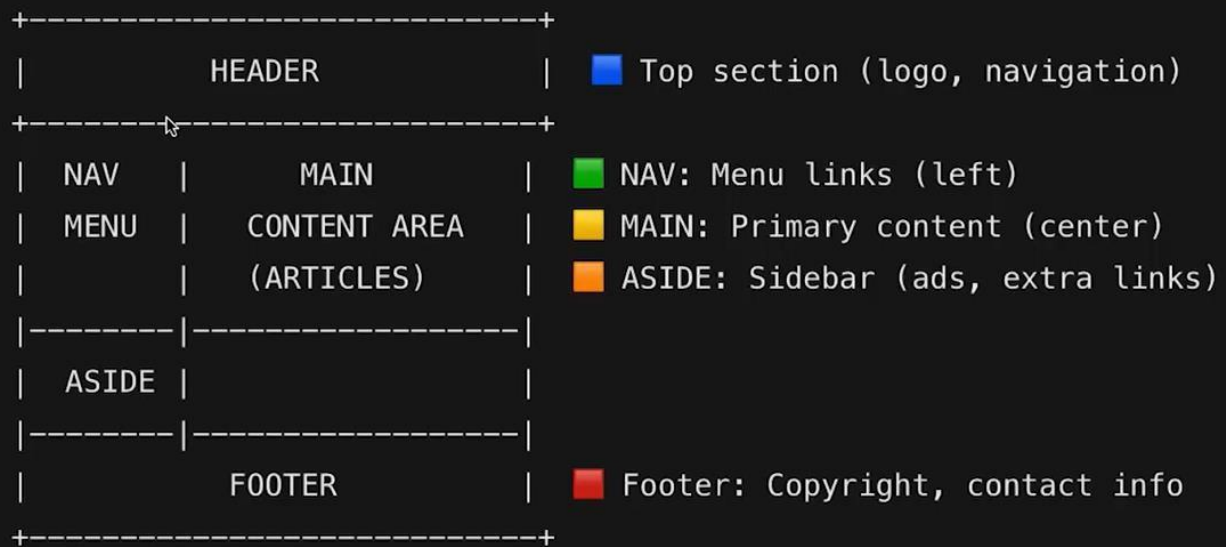
Semantic HTML



Common Semantic Tags

Tag	Purpose
<code><header></code>	Represents the top section of a page (logo, navigation).
<code><nav></code>	Contains navigation links (menus, sidebars).
<code><main></code>	The primary content of a page.
<code><section></code>	Groups related content (e.g., different parts of an article).
<code><article></code>	Represents self-contained content (e.g., blog post, news).
<code><aside></code>	Contains side content (ads, widgets, related links).
<code><footer></code>	The bottom section of a page (copyright, contact info).
<code><figure></code>	Wraps an image, diagram, or illustration.
<code><figcaption></code>	Provides a caption for a <code><figure></code> .

Visual Representation of Webpage Layout



Non-Semantic HTML (Messy Code)

```
<div id="header">
  <h1>Welcome to My Blog</h1>
</div>
<div id="nav">
  <a href="#">Home</a>
  <a href="#">About</a>
</div>
<div id="content">
  <p>This is a blog post.</p>
</div>
```

Semantic HTML (Clean Code)

```
<header>
  <h1>Welcome to My Blog</h1>
</header>
<nav>
  <a href="#">Home</a>
  <a href="#">About</a>
</nav>
<main>
  <article>
    <p>This is a blog post.</p>
  </article>
</main>
```

LocalStorage (The "Persistent Locker")

LocalStorage is a feature that allows you to store data directly in the browser.

Persistence: Unlike variables in JavaScript, data in LocalStorage does not disappear when you refresh the page or close the browser.

Use Case: It is perfect for your ShopEase project to remember what a user added to their cart even after they leave the site.

Capacity: You can store much more data (**about 5MB**) compared to older methods like Cookies.

```
// To save a product to the cart
localStorage.setItem('cartItem', 'Cool Shoes');

// To get that product back later
let item = localStorage.getItem('cartItem');
```

Script Loading (The "Traffic Control")

When adding JavaScript to your HTML, you need to decide how the browser should load it to keep the site fast.

Normal Script: The browser stops reading the HTML, downloads the script, runs it, and then continues. This makes the site feel slow.

async (Run as soon as ready): The script downloads in the background, but as soon as it's finished, it pauses the HTML to run.

defer (Wait for the Skeleton): This is the best practice. The script downloads in the background and only runs after the entire HTML structure (the skeleton) is fully loaded.

```
<script src="app.js" defer></script>
```

Form Validation (The "Security Guard")

Beyond just creating a <form>, you need to ensure users provide the right information.

required: Prevents the user from submitting the form if the input is empty.

type="email": Automatically checks if the user entered a valid email address format.

minlength: Forces the user to type at least a specific number of characters (useful for passwords).

```
<form>
  <input type="email" placeholder="Your Email" required>
  <input type="password" minlength="8" placeholder="Password"
                                             required>
  <button>Join ShopEase</button>
</form>
```

Media Tags (The "Cinema Hall")

Modern HTML lets you embed more than just images; you can add videos and audio directly.

<video>: Used to play movie files. Always include the controls attribute so users can press Play/Pause.

<audio>: Used for sound files or music.

alt & text: Just like alt for images, media tags help with accessibility.

```
<video src="promo.mp4" controls width="400">  
    Your browser does not support the video tag.  
</video>
```

Advanced SEO: Meta Tags (The "Signboard")

Search Engine Optimization (SEO) helps your shop appear higher on Google.

<meta name="description">: A short summary of your page that Google shows in search results.

<meta name="viewport">: Tells the browser how to scale your website so it looks good on mobile phones.

<title>: The main name of your page that appears on the browser tab.

```
<head>  
    <title>ShopEase - Best Shoes Online</title>  
    <meta name="description" content="Find the most comfortable  
        shoes at ShopEase.">  
    <meta name="viewport" content="width=device-width, initial  
        scale=1.0">  
</head>
```

What is HTML?

It is the skeleton of a website. It builds the structure.

What are HTML Tags?

They are labels (like `<h1>`) that tell the browser what to show.

What are Attributes?

They provide extra info about a tag (like `src` for an image).

What is `<!DOCTYPE html>`?

It's a signal telling the browser: "Hey, I am using HTML5!"

Difference between head and body?

Head: Hidden info (Brain). Body: What you see (Face).

Difference between div and span?

Div: A big box (Block). Span: A small highlighter (Inline).

What are Void Elements?

Tags that don't need a closing tag, like `<img>` or `<br>`.

What is an Anchor tag?

The `<a>` tag used to create links to other pages.

What is an Iframe?

A window to show another website inside your website.

Difference between id and class?

ID: Unique (1 person). Class: Group (a whole team).

What are Semantic Tags?

Labeled boxes like `<header>` and `<footer>` that explain their purpose.

What is the alt attribute?

A text description for an image if it fails to load.

How to open a link in a new tab?

Use the attribute `target="_blank"`.

What is a `<form>`?

An envelope used to collect and send user data to the server.

What is the type attribute in input?

It defines the input style (e.g., text, password, or checkbox).

What are Meta Tags?

Hidden info inside the `<head>` that helps Google find your site.

What is SVG?

Scalable images that never get blurry when you zoom in.

Difference between `<ul>` and `<ol>`?

UL: Bullet points (Dots). OL: Numbered list (1, 2, 3).

What is a Favicon?

The tiny icon you see on the browser tab.

Why is SEO important in HTML?

It helps your website appear higher in Google search results.

Difference between `LocalStorage` and `SessionStorage`?

LocalStorage: Data stays in the browser forever (until you delete it).

SessionStorage: Data is deleted as soon as you close the browser tab.

Difference between `async` and `defer` in script tags?

async: Downloads the script and runs it immediately, pausing the HTML.

defer: Downloads the script but waits until the entire HTML (the skeleton) is finished before running. (Best for performance).

Why should we use the <label> tag for inputs?

It is for Accessibility. It helps screen readers identify the input, and if you click the label text, it automatically selects the input box.

When should you use a <a> tag vs a <button>?

<a> (Anchor): Use this when you want to navigate to a different page or section.

<button>: Use this when you want to perform an action (like "Add to Cart" or "Submit Form").

What are data-* attributes?

They allow you to store extra information inside an HTML tag that you can easily grab using JavaScript later.

```
<div data-product-id="101">Nike Shoes</div>
```

What is the purpose of the <meta name="viewport"> tag?

It ensures your website is Responsive. It tells the browser to scale the website width to fit the screen size of the device (Mobile, Tablet, or Desktop).